INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME

CLASS

INDEX NUMBER

BIOLOGY 9648/01

Paper 1 Multiple Choice

24 September 2014

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

This document consists of **18** printed pages.

1 Which cell components contain ribosomes?

1 chloroplast

2 cytoplasm

3 Golgi apparatus

4 mitochondrion

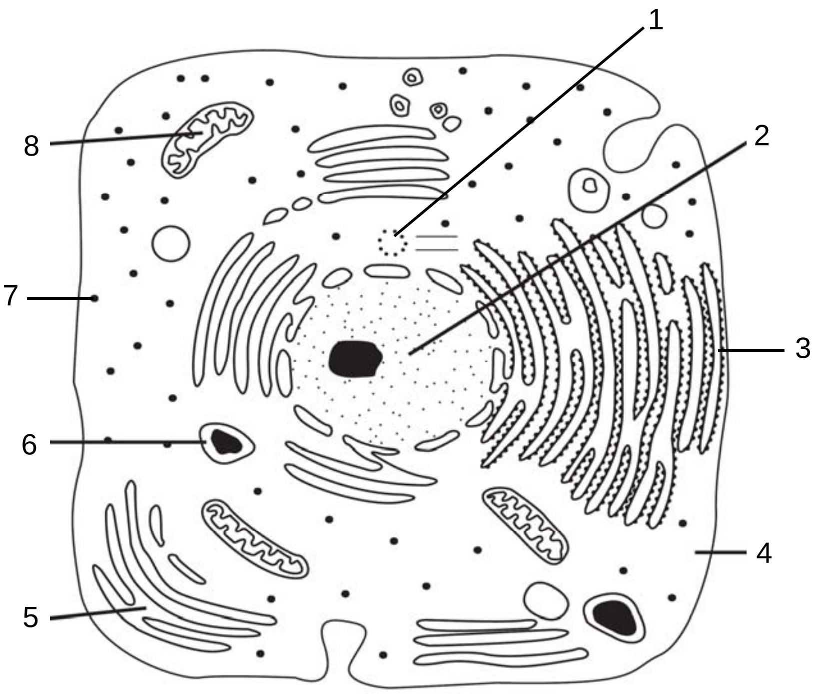
A 1 and 2 only

B 1 and 4 only

C 1, 2 and 4 only

D 1, 2, 3 and 4

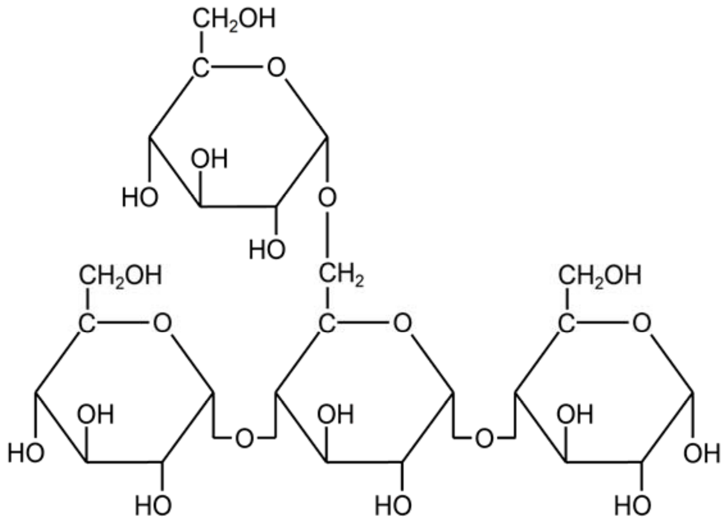
2 The diagram shows a drawing of an electron micrograph of an animal cell.



Which of the following describes the corresponding properties of the labelled structures?

	contains nucleic acids	undergoes doubling during cell division	contain enzymes
A	2, 8	2	6, 8
B	2, 3, 8	2, 4, 8	5, 6, 8
C	1, 2, 7, 8	1, 2, 8	2, 4, 6, 8
D	2, 3, 7, 8	1, 2	2, 3, 4, 5, 6

3 The diagram shows a carbohydrate molecule.



From which of the following polymers could this be a part of?

1 amylopectin
2 amylose
3 cellulose
4 glycogen

A 1 only
B 3 only
C 1 and 2 only
D 1 and 4 only

4 Which of the following bonds maintains the α-helix structure in proteins?

1 hydrogen bonds between peptide bonds
2 hydrogen bonds between polar R groups
3 hydrophobic interactions between non-polar R groups
4 ionic bonds between oppositely charged R groups
5 peptide bonds between amine and carboxyl groups

A 1 only
B 1 and 2 only
C 1, 2 and 5 only
D 1, 2, 3 and 4

5 Which of the following statement is true for **all** enzymes?

A They alter the activation energy needed for a reaction.

B They break down complex substrates into simpler products.

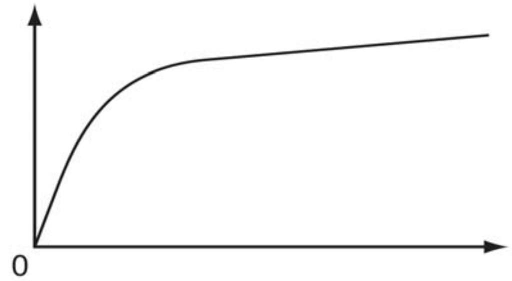
C They can increase the speed of endothermic reactions.

D They denature at temperatures greater than 60^{o}C.

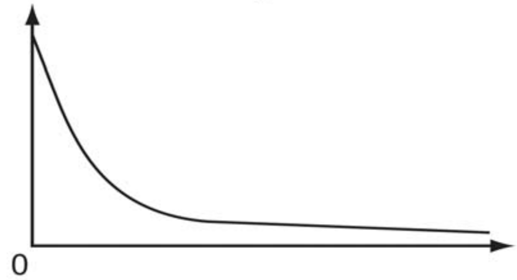
6 In an experiment, 5 cm^3 of 1 % salivary amylase are added to 100 cm^3 of 5% starch.

Which graph shows the correct results of plotting the rate of reaction (y-axis) against the time (x-axis)?

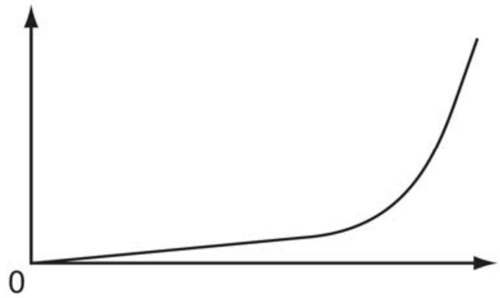
A



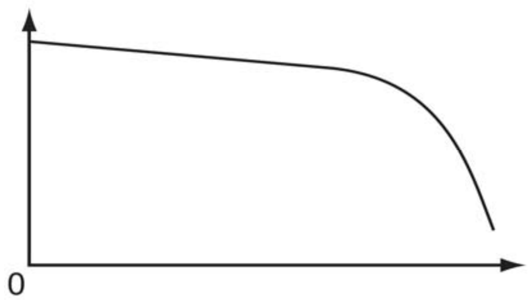
B



C



D



7 A cell with 10 chromosomes at G_1 phase divides. How many chromosomes will each cell have at different stages of the cell cycle?

	prophase of mitosis	end of mitosis	end of meiosis I
A	5	10	5
B	10	10	5
C	20	10	10
D	20	10	20

8 Which of the following statement is true about DNA polymerase?

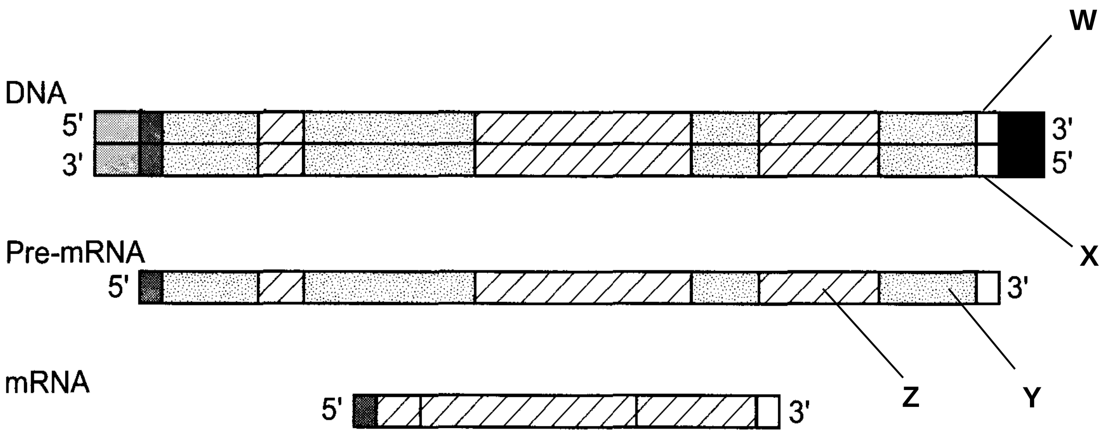
A It synthesizes the leading strand from 5' → 3' and the lagging strand from 3' → 5'.

B It synthesizes a polynucleotide using the non-sense strand as the template.

C It synthesizes a RNA using DNA as a template.

D It synthesizes a strand of DNA using DNA as a template.

9 The diagram shows the processing of transcribed pre-mRNA in a eukaryotic cell.



Which of the following identifies structures **W** to **Z**?

	W	X	Y	Z
A	non-template strand	template strand	exon	intron
B	non-template strand	template strand	intron	exon
C	template strand	non-template strand	intron	exon
D	template strand	non-template strand	exon	intron

10 Which of the following statement is true about silent mutation?

A Silent mutations may be result of degeneracy in the genetic code.

B Neutral mutations are silent mutations.

C Single base deletions in introns are silent mutations.

D Single base substitution at the third base of a codon will result in silent mutations.

11 Which of the following statement, comparing the influenza virus and lambda phage, is **incorrect**?

A The influenza virus infects mammalian cells, but the lambda phage infects bacterial cells.

B The capsid of the influenza virus enters the host cell, but the capsid of the lambda virus does not.

C The DNA genome of the influenza virus is not integrated into the host cell's genome, but the DNA genome of the lambda virus is.

D New influenza virions are released from the host cell via budding, but new lambda virions are released via cell lysis.

12 Shingles is a painful skin disease caused by the reactivation of the chickenpox virus (*Varicella-zoster* virus) and it occurs more commonly in the elderly. Scientists have developed a vaccine which comprises a live, weakened form of the virus. The vaccine triggers a mild immune response resulting in antibodies production.

Which stage of the viral reproductive cycle is directly affected after an individual is vaccinated against shingles?

A Attachment stage

B Viral synthesis stage

C Assembly stage

D Release stage

13 Which of the following correctly describes a prokaryotic genome?

	Presence of homologous chromosomes	Presence of histone proteins	Presence of DNA in organelles	Presence of plasmids
A	✓	✓	✓	×
B	×	✓	✓	×
C	✓	×	×	✓
D	×	×	×	✓

14 In an inducible operon under negative control, two mutations were found to have occurred:

1 Mutation 1 renders the product of the operon's regulatory gene non-functional.

2 Mutation 2 leads to a missense mutation in its structural genes.

These mutations will result in

A continuous transcription of genes of the operon.

B no transcription of genes of the operon.

C irreversible binding of the repressor to the operon.

D no difference in the transcription rate.

15 Which of the following mechanism may occur before eukaryotic cells begin transcription?

A DNA acetylation

B DNA methylation

C histone acetylation

D histone methylation

16 Which of the following is **not** a control element?

A enhancer

B silencer

C promoter

D repressor

17 Which of the following is **not** a requirement for the mechanism of RNA splicing?

A Formation of a pre-mRNA-snRNP complex

B Involvement of guanylyl transferase

C 5’ and 3’ splice sites

D Complementary base pairing between snRNAs with conserved sequences

18 The length of the wing in a particular species of moth is under the genetic control of two genes. The population of moths displays three distinct phenotypes: vestigial, normal and long. The two genes involved were found to be autosomally linked.

Which of the following correctly describes the features of genes displaying autosomal linkage?

1 The genes assort independently.

2 The genes are on the same chromosome.

3 There may be recombination between the two genes during meiosis.

4 Particular combinations of alleles of these genes tend to be inherited together.

A 1 and 2

B 2 and 3

C 1, 2 and 3

D 2, 3 and 4

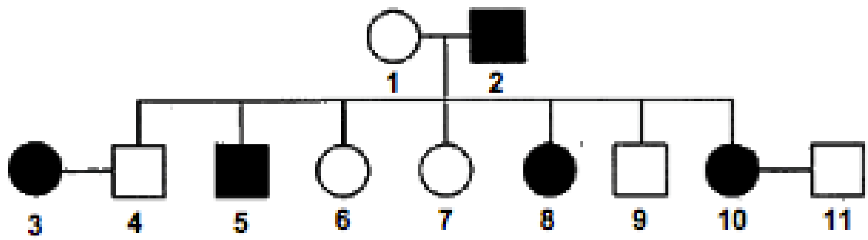
19 A plant is heterozygous for two genes, **A** and **B**. Alleles **A/a** display codominance while alleles **B/b** display a dominant-recessive relationship. The plant is subjected to a test cross and the resulting seeds were germinated and allowed to grow.

Which of the following ratios are expected in the offspring?

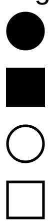
	ratio of phenotypes	ratio of genotypes
A	1 : 1 : 1 : 1	1 : 1 : 1 : 1
B	1 : 1 : 1 : 1	1 : 2 : 1
C	1 : 2 : 1	1 : 1 : 1 : 1
D	1 : 2 : 1	1 : 2 : 1

20 Night blindness is a condition in which affected people have difficulty seeing in dim light. The allele N for night blindness is dominant to the allele n for normal vision.

The diagram below shows part of a family tree illustrating the inheritance of night blindness.



Legend



female with night blindness

male with night blindness

female with normal vision

male with normal vision

If the first child born to individuals 10 and 11 is a boy with night blindness, what is the probability that the next child being a boy with night blindness?

A 0.0625

B 0.125

C 0.25

D 0.50

21 The diagram below shows the steps in a metabolic pathway leading to the production of a brown pigment in the fur of cats.

colourless substrate $\xrightarrow{\text{enzyme } \mathbf{G}}$ colourless intermediate $\xrightarrow{\text{enzyme } \mathbf{B}}$ brown pigment

The gene **G** codes for enzyme **G** and gene **B** codes for enzyme **B**. The genes are at different loci on non-homologous chromosomes.

A cross between two white-fur cats produced offspring, all with brown fur.

What were the genotypes of the white-fur cats?

A **GGbb** and **ggBB**

B **Ggbb** and **ggBb**

C **Ggbb** and **ggBB**

D **ggbb** and **GGbb**

22 A blue bird with red eyes with the genotype **BBrr** was crossed with a white bird with black eyes with the genotype **bbRR**. The F_1 birds were allowed to cross-fertilised. A chi-squared test was carried out on the results obtained for the F_2 generation which showed the phenotypic ratio of 12:3. Part of the table of values for χ^2 is shown.

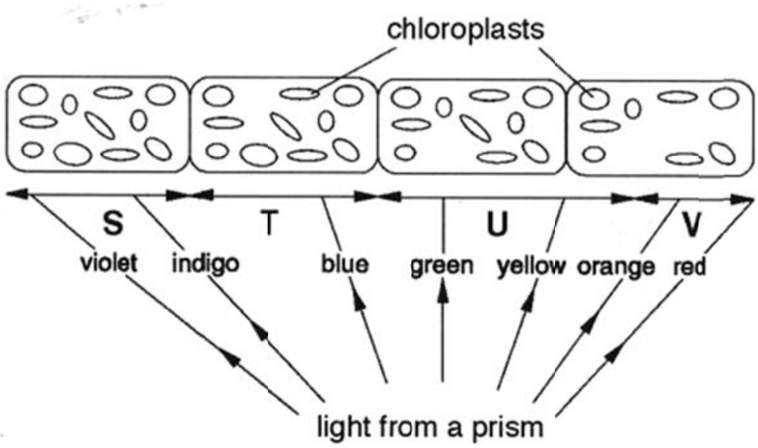
degree of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

The value χ^2 in this investigation was 6.36.

Which of the following conclusion made about the inheritance of bird plumage colour and eye colour is correct?

	probability, p	difference between observed and expected results
A	between 0.01 and 0.05	not significant
B	between 0.01 and 0.05	significant
C	between 0.02 and 0.05	not significant
D	between 0.02 and 0.05	significant

23 A filament of the alga *Cladophora* was illuminated as shown in the diagram. Motile, aerobic bacteria were placed in the water with the alga and their position was determined by microscopic examination after 10 minutes.



In which two regions will the number of bacteria be highest?

A S and T

B S and U

C T and V

D U and V

24 In a series of experiments, actively photosynthesising plants were supplied with radioactively-labelled water containing ^{18}O isotope and carbon dioxide containing ^{13}C isotope.

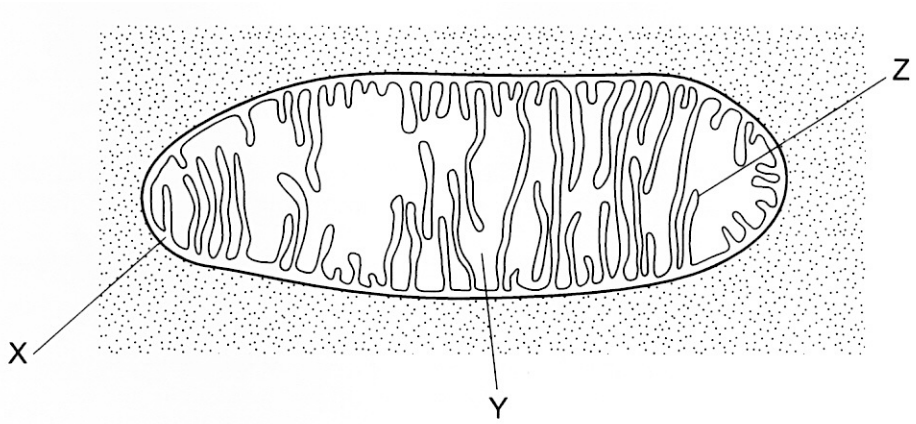
Where, in chloroplast, would the products of photosynthesis from these reactants be formed?

	^{18}O	^{13}C
A	thylakoids	stroma
B	thylakoids	thylakoids
C	stroma	thylakoids
D	stroma	stroma

25 At which stage in the oxidation of glucose are both ADP and NAD required?

$$\text{glucose} \xrightarrow{\textbf{A}} \text{hexose phosphate} \xrightarrow{\textbf{B}} \text{triose phosphate} \xrightarrow{\textbf{C}} \text{pyruvate} \xrightarrow{\textbf{D}} \text{acetylcoA}$$

26 The electron micrograph shows an organelle found in a plant cell.



Which row shows the correct location of each named biological molecule?

	DNA	oxaloacetate	ATP
A	absent	Y	Z
B	Y	Z	Y
C	Y	Y	X
D	Y	Y	Y

27 Transmission of action potential along the axon is unidirectional because

A Na^+ only diffuse in one direction.

B the myelin sheath causes the potential to jump from one node of Ranvier to the next.

C the preceding region is undergoing repolarization.

D of the local current that is set up.

28 Adrenaline binds to a receptor on a cell surface membrane and, as a result, the activity of an enzyme, E, in the cell is altered by phosphorylation.

Some statements about this example of cell signalling are listed.

1 The concentration of cyclic AMP (cAMP) in the cell increases.

2 A kinase enzyme adds a phosphate group to its substrate.

3 Cyclic AMP (cAMP) activates an enzyme.

4 The enzyme adenylyl cyclase is activated.

Which sequence of steps occurs to alter the activity of enzyme E?

A 1 → 4 → 2 → 3

B 2 → 3 → 4 → 1

C 3 → 2 → 1 → 4

D 4 → 1 → 3 → 2

29 Which taxa do *Ursus arctos* and *Ursus maritimus* share?

A They share the same class but not the same family.

B They share the same genus but not the same class.

C They share the same species but not the same class.

D They share the same family but not the same species.

30 Which of the following are parts of Darwin's theory of evolution by natural selection?

1 Variation between individuals arises by gene mutation.

2 Acquired trait is inherited by the offspring.

3 Organisms tend to produce more offspring than the environment can support.

4 Only those individuals best adapted to the environment survive and reproduce.

A 1 and 2

B 3 and 4

C 1, 3 and 4

D 1, 2, 3 and 4

31 Which of the following would provide the best information for distinguishing phylogenetic relationships between several species that are almost identical in anatomy?

A Fossil records

B Homologous structures

C Comparative embryology

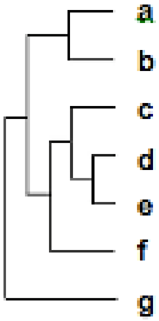
D Molecular comparison of DNA and amino acid sequences

32 The table below shows the number of estimated nucleotide substitutions that have occurred since the divergence of seven species **a** to **g**.

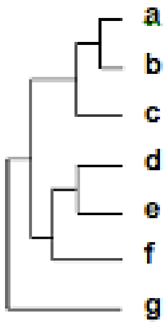
	a	**b**	**c**	**d**	**e**	**f**
b	16					
c	56	51				
d	66	64	34			
e	61	68	35	19		
f	245	221	234	238	226	
g	237	224	229	242	231	13

Which of the following phylogenetic trees best shows the relationship among these seven species?

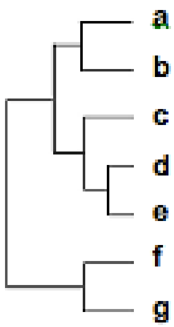
A



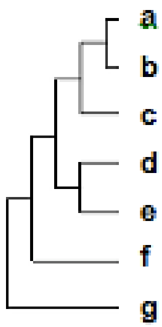
B



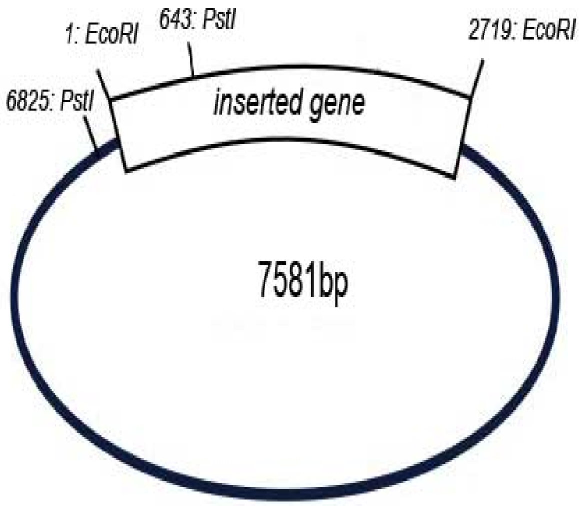
C



D



33 A scientist planned to insert a gene of approximately 2719 base pairs into a plasmid vector using restriction enzyme *Eco*RI. The diagram below shows the recombinant plasmid with the gene in the desired orientation.

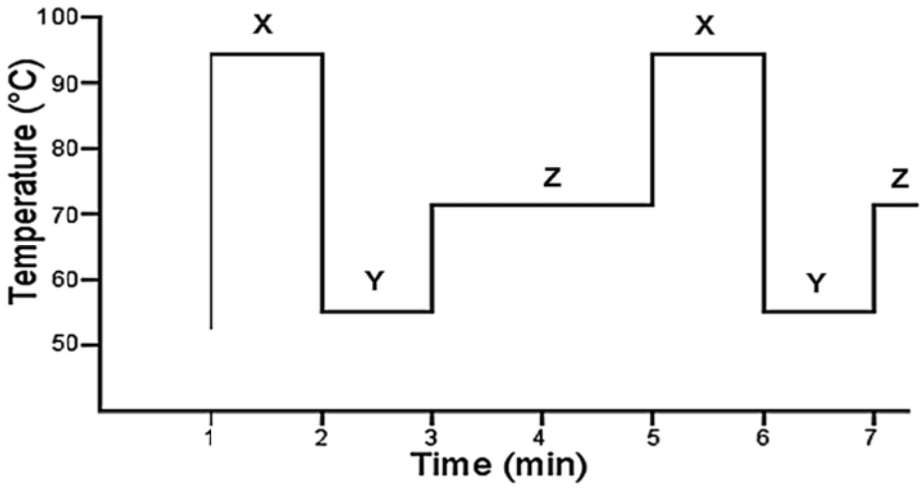


Restriction mapping can be used to check the orientation of the gene inserted into a plasmid. The recombinant plasmid is cleaved with restriction enzymes(s) and the fragments are separated on an electrophoresis gel to determine their size.

Which of the following results will confirm that the gene was **not** inserted in the desired orientation?

	Restriction enzyme(s) used	Fragment size(s)
A	*Pst*I	1399bp, 6182bp
B	*Pst*I	2832bp, 4749bp
C	*Eco*RI	2719bp, 4862bp
D	*Pst*I and *Eco*RI	643bp, 756bp, 2076bp, 4106bp

34 The diagram below shows the changes in temperature in a thermal cycler over time during Polymerase Chain Reaction.



Which of the following statement is true?

A DNA primers are anneal to the DNA template during **Z**.

B Double stranded DNA template denatures into single strands during **X**.

C Elongation of new daughter strand occurs during **Y**.

D *Taq* polymerase functions optimally at **X**.

35 The Human Genome Project (HGP) has brought about great advancements in health and medicine.

Which of the following applications is **not** a result of the knowledge gained from the HGP?

A Designing of new antibody-based medicines which target proteins coded for by oncogenes.

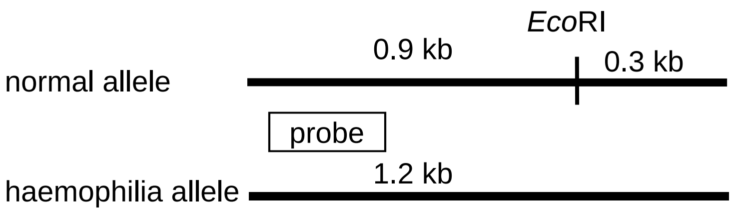
B Determining whether an individual is a suitable candidate for working in a nuclear power plant due to a genetic predisposition to cancer.

C Predicting whether an individual may suffer adverse side effects from taking a particular medicine due to the individual's inability to break down the medicine.

D Testing the gender of a foetus by detecting the presence or absence of the fetal Y chromosome in maternal blood.

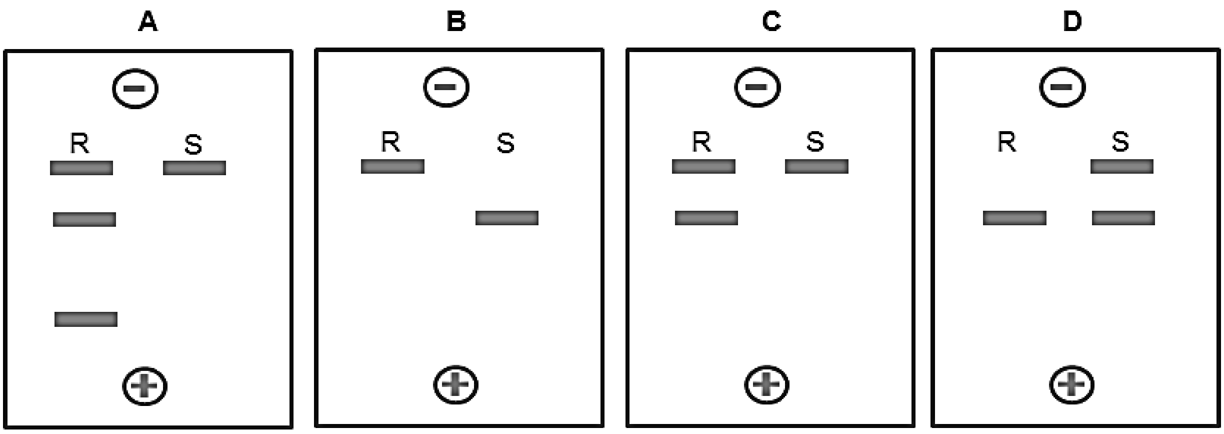
36 Haemophilia A is a X-linked disease resulting from a mutation in the gene that codes for Factor VIII, a protein needed for blood clotting. The haemophilia allele is recessive to the normal allele.

The diagram show the position in the normal and haemophilia allele which the radioactive probe binds.



A man who suffers from the disease married a woman who is a carrier of the disease. The couple has 2 children. Child **R** is normal while Child **S** is affected.

Which autoradiograph correctly identifies the children?

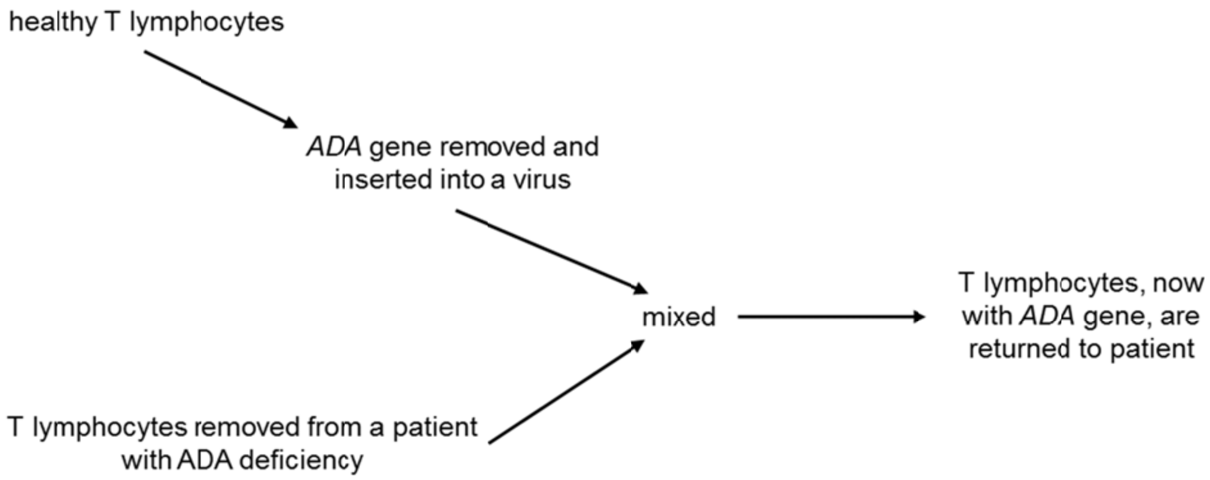


37 Which of the following statements are true about adult stem cells?

1 They can undergo self-renewal.
2 They can be totipotent, pluripotent or multipotent.
3 They can differentiate into almost any cell type.
4 They can give rise to specialised cells.

A 1 and 2
B 1 and 4
C 2 and 3
D 3 and 4

38 The diagram shows a gene therapy approach for SCID.



Which of the following factors prevent this gene therapy approach from becoming an effective treatment for the patient?

1 The patient may develop leukaemia.

2 The virus may lyse the target T lymphocyte.

3 The normal ADA allele may be passed to the next generation.

4 This approach may not be permanent, thus repeated gene therapy is necessary.

A 1 and 4 only

B 2 and 3 only

C 1, 2 and 4 only

D 1, 3 and 4 only

39 Which of the following correctly matches the plant tissue culture techniques to their main purpose?

	anther culture	protoplast culture	suspension culture	callus culture
A	to produce hybrid plants	to produce genetically identical plants	to screen for recessive desirable traits	to extract useful secondary metabolites
B	to screen for recessive desirable traits	to produce hybrid plants	to extract useful secondary metabolites	to produce genetically identical plants
C	to produce plantlets which are genetically non-identical	to produce transgenic plants	to trigger shoot and root formation	to produce disease free plants
D	to produce disease free plants	to trigger shoot and root formation	to produce transgenic plants	to produce plantlets which are genetically non-identical

40 Which row correctly states the effects and possible consequences of producing the various genetically-modified organisms?

	Bt corn		GM salmon		Golden rice	
	effect	possible consequence	effect	possible consequence	effect	possible consequence
A	improve quality	might become weeds	improve yield	displace native species	improve yield	might become weeds
B	improve yield	toxic to closely related species	improve yield	displace native species	improve quality	cause allergies
C	improve yield	might become weeds	improve quality	cause allergies	improve yield	cause allergies
D	improve yield	toxic to closely related species	improve yield	cause allergies	improve quality	might become weeds

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2014 JC2 Prelim 2
9648 H2 Biology
Paper 1
Suggested Answers

Qn	Ans	Qn	Ans	Qn	Ans	Qn	Ans
1	C	11	C	21	A	31	D
2	D	12	A	22	B	32	C
3	D	13	D	23	C	33	B
4	A	14	A	24	A	34	B
5	A	15	C	25	C	35	D
6	B	16	D	26	D	36	C
7	B	17	B	27	C	37	B
8	D	18	D	28	D	38	A
9	B	19	A	29	D	39	B
10	A	20	C	30	B	40	B